"Python Project Documentation"

Chatbot for the subject: [Integrated Programming and Tech]

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1. Introduction

This project develops a chatbot for Integrated Programming and Technology, designed to assist students in understanding how multiple technologies can be combined to solve real-world problems. The chatbot provides accurate, subject-specific answers in a conversational format, demonstrating integration of programming, web technologies, and artificial intelligence.

Purpose: To showcase the integration of programming concepts with modern AI tools.
Intended Users: BSCSIT students and faculty members evaluating the project.

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2. Development Process

Platform and Tools Used

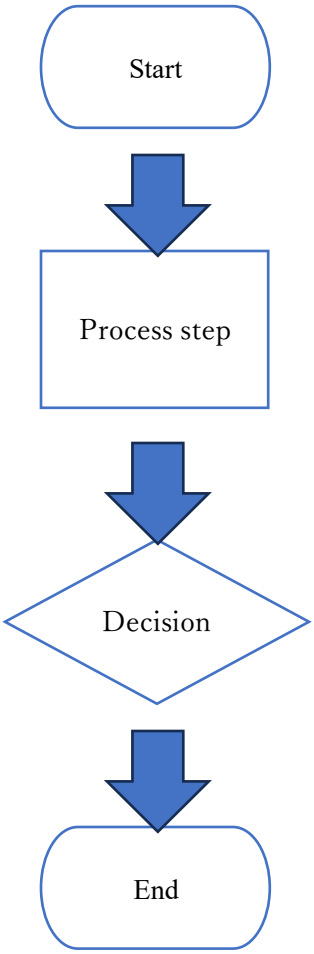
- Python 3.13
- Flask (for web interface)
- Google Gemini API (for AI responses)
- Local network deployment (IP: 10.1.0.40)

Design and Planning

The following flowchart illustrates the chatbot's process:

1. User inputs a question.
2. Flask processes the request.
3. Gemini API generates a response.
4. The response is displayed in the browser.

Flowchart image



Implementation

Key code snippet: python

```
@app.route('/chat', methods=['POST'])
def chat():
    data = request.get_json()
    user_message = data.get("message", "")
    response = model.generate_content(user_message)
    return jsonify({"reply": response.text})
```

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3. Features

- User Input:

  Text-based questions via browser.

- Response Generation:

  AI-powered answers using Gemini API.

- Unique Feature:

  Replies are generated in poetry format, making learning engaging.

- Example: Input:

  "What is a loop?" Output: "A loop is like a wheel that spins, / Repeating code until it wins..."

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4. Deployment Details

- **Local Deployment:**

  Run python app.py → Server starts at http://10.1.0.40:5000

- **Access Details:**

  Professor can access the chatbot on November 19, 2025 via:
  http://10.1.0.40:5000

- **Network Setup:**

  - IP fixed to 10.1.0.40

  - Firewall configured to allow TCP port 5000

5. Challenges and Solutions

API key security → Generated a new key and kept it private

IP configuration → Manually set static IP (10.1.0.40)

Firewall blocking → Allowed port 5000 in Windows Defender

Response length → Limited output tokens to avoid cutoff

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6. Conclusion and Learnings

The chatbot successfully demonstrates the integration of programming and technology by combining Python, Flask, AI APIs, and networking concepts. It is accessible online and responds accurately to subject-related queries.
Learnings:

・How to integrate multiple technologies into one system

・Importance of deployment and network configuration

・Documentation and project presentation skills

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7. References

・Google Generative AI Documentation

https://ai.google.dev

・Flask Documentation

https://flask.palletsprojects.com